TRƯỜNG ĐẠI HỌC CÔNG NGHỆ KỸ THUẬT TP.HCM
KHOA ĐIỆN-ĐIỆN TỬ



PROJECT

RESEARCH, DESIGN, AND DEVELOPMENT OF A QUADCOPTER SYSTEM FOR AUTOMATED DROWNING RESCUE OPERATIONS

Supervising teacher: Associate Professor. LÊ MỸ HÀ, Ph.D

Student: THAKORN TANGPHADUNGRUTH

TP. HỒ CHÍ MINH, February 2026

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LIST OF ABBREVIATIONS

Abbreviations	Content
SC	Invention
University	University
Ho Chi Minh City	Ho Chi Minh City
TN	Experiment

UAV	Unmanned Aerial Vehicle
GPS	Global Positioning System
AI	Artificial intelligence
CW	Clockwise
CCW	Counterclockwise
DOF	Degrees of Freedom
PID	Proportional Integral Derivative
NOSE	Inertial Measurement Unit
GCS	Ground Control Station
ECF	Extended Kalman Filter

CHAPTER 1: OVERVIEW

This chapter synthesizes and analyzes the research landscape on UAV systems for drowning rescue, both domestically and internationally, to highlight the urgency and practical importance of this research direction. It then outlines the specific objectives, scope, and limitations of the study to clearly define the methodological approach. Additionally, the chapter discusses the research methods employed to ensure scientific rigor and logical progression throughout the investigation. Finally, it provides an overview of the main content to guide the development of subsequent chapters.

1.1 Problem statement

In Vietnam's vast coastline, rivers, and flood-prone areas, drowning is a major public health crisis, claiming about 2,000 children's lives annually—the top cause of death for those under 15—due to poor swimming education, supervision gaps, and access challenges in remote regions. In 2021, over 4,000 total drowning deaths occurred, with 40% affecting children under 14, underscoring the need for innovative rescues in urban hubs like Hanoi and Ho Chi Minh City, and vulnerable provinces.

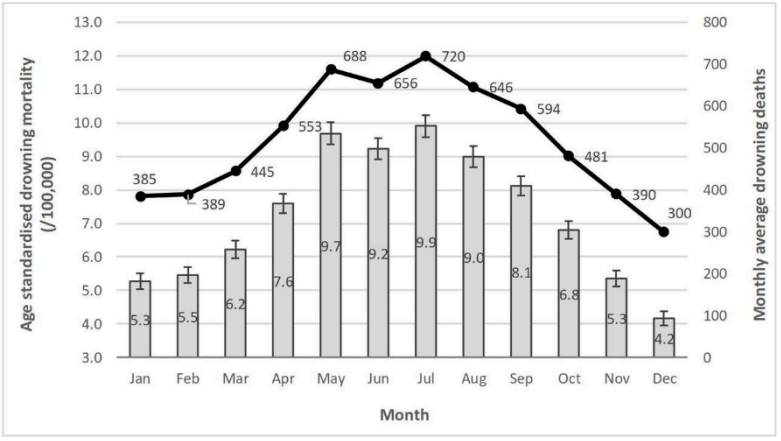


Figure 1. Statistics on drowing deaths in Vietnam

Globally, drowning kills around 300,000 people yearly, ranking third in unintentional injury deaths, with 90% in low- and middle-income countries. Over half strike the WHO Western Pacific and South-East Asia regions, where kids under 5 and 5-14 account for 24% and 19% of cases. Despite a 38% rate drop since 2000 (from 6.1 to 3.8 per 100,000), it remains preventable, tied to urbanization, climate risks, and weak infrastructure. Advanced nations innovate with drones, like Microdrones systems in Australia and the US, for delivering flotation aids, surveillance, and safer, faster

responses.

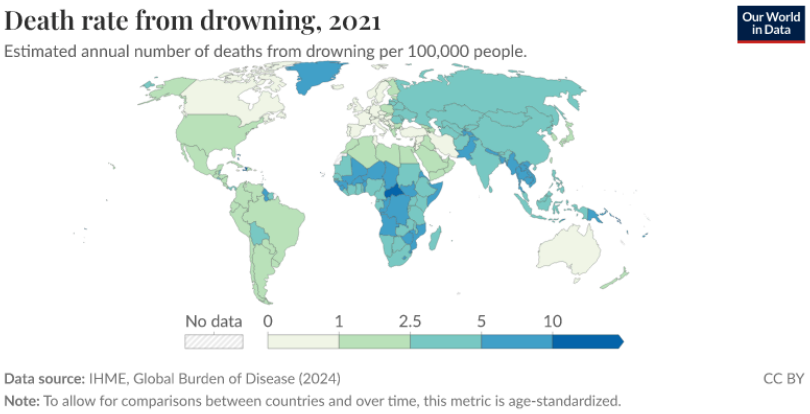


Figure 2. Statistics on drowing deaths worldwide

In Vietnam, drones hold great potential amid 3,260 km of coastline and river networks, but face hurdles like limited payload, battery life, weather safety risks, and incomplete regulations. Research on victim detection algorithms, energy optimization, and autonomy is limited, hindering local adaptation of global tech.

Economically, drones cut costs by reducing human teams and injury-related healthcare loads. Socially, they enable quick responses in high-risk zones, boosting child safety and disaster readiness per national goals. Environmentally, they provide low-emission alternatives to boats or helicopters, aiding sustainability in a climate-vulnerable nation.

Thus, the study "Research, Design, and Development of a Quadcopter System for Automated Drowning Rescue Operations" designs an optimized drone, analyzes techniques, builds a prototype, and compares it to traditional methods. Focused on coastal and urban rivers, it weighs response time, costs, and safety, using simulations (Python, ROS) and real tests for feasibility, aiming to advance smart rescue systems in Vietnam.

1.1.1 The situation of overseas research

Unmanned Aerial Vehicles (UAVs) are aircraft that operate without a direct pilot on board, controlled remotely or autonomously through integrated computer systems. Due to their advantages in safety, flexibility, and the ability to integrate electronic devices, cargo, or equipment, UAVs have become a significant focus of global research. Initially designed for military purposes, reconnaissance, and rescue

operations, UAVs have now expanded into civilian applications. Among various UAV types, the Quadcopter—a four-rotor aircraft with an X or cross-shaped frame—is the most common design, generating thrust through aerodynamic effects, offering flexible operation, ease of control, and reasonable costs.

The rapid advancement of Industry 4.0 technologies and computer vision, particularly in machine learning, has boosted the potential applications of UAVs. Equipped with sensors and embedded computers, UAVs can perform high-precision tasks such as target detection and tracking via imaging. The integration of cameras and processors on Quadcopters opens up practical solutions in everyday life, including automated drowning rescue operations.

Internationally, research and applications of UAVs in drowning rescue have made significant strides, particularly with the support of global positioning systems (GPS) and autonomous control technologies. Leading initiatives include Australia's Little Ripper drone, which has been deployed for beach surveillance and rapid delivery of flotation devices to distressed swimmers, successfully saving lives in real-world scenarios. In the United States and Europe, companies like Microdrones have developed systems such as the mdSAR (Microdrones Search and Rescue), equipped with cameras and compact flotation devices like RESTUBE, enabling drones to reach drowning victims in as little as 30 seconds—far faster than traditional lifeguard responses—and reducing risks to rescuers. Studies in Spain, such as the Auxdron Lifeguard project, integrate LiDAR sensors and computer vision for obstacle avoidance and precise victim detection, achieving up to 90% accuracy in landing flotation aids even in crowded or urban aquatic environments. In China and Poland, research focuses on IoT-integrated drones for monitoring and rescue, using ArUco markers for precise positioning and infrared LEDs to overcome lighting and GPS dependencies, enhancing safety in adverse weather. However, challenges such as payload limitations, battery life, and regulatory frameworks are being addressed through advancements in AI-driven algorithms for victim recognition and energy management. These innovations not only improve rescue efficiency—halving response times compared to conventional methods—but also

align with "Smart City" strategies for sustainable emergency response.

1.1.2 The state of domestic research

Vietnam is experiencing a high incidence of drowning incidents, with over 4,000 deaths annually, particularly in coastal and riverine areas, driven by factors such as limited swimming education, seasonal flooding, and inadequate rescue infrastructure in urban centers like Hanoi and Ho Chi Minh City. However, emergency response systems have not kept pace with the demand, leading to delayed interventions and high logistics costs for rescue operations, which account for significant public health expenditures. In this context, drones for drowning rescue are viewed as a promising solution to shorten response times, especially in densely populated or remote regions. Nevertheless, the application of this technology in Vietnam remains limited, primarily due to incomplete regulations on airspace management, technological barriers (such as payload capacity and battery life), and initial investment costs. Some pilot projects have been implemented, such as drone-assisted rescues during floods in Quang Ninh or support in mountainous areas, but most are on a small scale and not yet commercialized.

To advance drone applications in drowning rescue, Vietnam must address a series of challenges, from establishing clear legal frameworks for airspace zoning and data security to enhancing domestic technological capabilities. Additionally, the tropical monsoon climate with unpredictable storms and heavy rains imposes requirements for weather-resistant drone designs. However, the potential is immense, especially since 64% of the population lives in rural areas with limited access via traditional means. Local conglomerates like Viettel and domestic startups are gradually researching drones integrated with AI and IoT, focusing on optimizing routing and energy management. The government has shown support through the National Project on the Fourth Industrial Revolution, paving the way for pilot initiatives. If successful, drones for drowning rescue could not only alleviate the burden on traditional rescue infrastructure but also promote green emergency responses, contributing to the modernization of Vietnam's public safety sector.

Research on unmanned aerial vehicles (UAVs) in Vietnam is in its early development stage, attracting particular interest from the student community. However, the field faces numerous challenges due to budget constraints and the lack of specialized training programs at universities. Currently, most student-led UAV projects remain at a basic level, although some notable topics have emerged:

At Da Nang University of Science and Technology, researchers have explored drone designs for rescue operations, providing simulation results and theoretical flight models for Quadcopters based on control systems. From 2018 to 2023, numerous studies on Quadcopters have been conducted at Vietnamese universities, demonstrating clear technological progress. In 2018, teams at Ho Chi Minh City University of Technology and Education focused on PID control for autonomous landing on fixed targets. By 2019, developments incorporated Arduino and human detection algorithms. In the 2022-2023 period, research shifted toward AI applications: One study in 2022 utilized Tello Drones and YOLO for object tracking, while another in 2023 emphasized automated rescue solutions with advanced navigation algorithms. Notably, a group in 2023 achieved a breakthrough by integrating Jetson Nano for real-time processing, offering high stability. The overall trend indicates a shift from traditional control to intelligent systems, despite ongoing challenges with hardware and training data.

Domestic Quadcopter research from 2018-2024 has primarily focused on theoretical simulations and basic applications like takeoff/landing, while real-world applications, particularly in image processing for drowning detection, remain limited. Recognizing this gap, the research team proposes the project "Research, Design, and Development of a Quadcopter System for Automated Drowning Rescue Operations," aiming to develop a cost-effective flying solution integrated with image processing technology to meet practical needs such as surveillance or rapid intervention. This approach not only addresses previous limitations but also opens up broad potential applications in the future.

1.2 Project objective

1.2.1. Purpose

Research, design and development of a quadcopter system for automated drowning resue operation.

1.2.2. Objectives

The objectives of the project are set as follows:

- Calculate the kinematic equations describing the unmanned aerial vehicle (UAV), thereby designing a flight model capable of stable flight.
- Develop a program to track GPS waypoints while the aircraft is in flight, and design software capable of monitoring and planning flight path trajectories.
- Apply UAV position control to maintain position in space and fly along trajectories based on GPS signals.
- Apply computer vision to identify the status of drowning individuals and send GPS coordinates to the monitoring software.

1.3 Topic limitations

- The unmanned aerial vehicle (UAV) is allowed to fly to a pre-selected GPS point, with no obstacles along the flight path.
- The maximum cargo weight that can be carried is limited to 5 kg.
- The maximum speed of the drone is 5 m/s.

1.4 Research methods

This project combines theoretical and experimental research methodologies.

Theoretical Research: The project conducts a comprehensive overview of unmanned aerial vehicle (UAV) technology for rescue operations, focusing on optimal flight patterns, hovering stability, and lifebuoy deployment methods over water. Additionally, it investigates UAV control systems and computer vision algorithms for object detection in aquatic environments to support system modeling.

Experimental Research: Upon completing the model, the team will begin manual flight tests using a remote controller to assess the aircraft's stability during takeoff, directional flight, and landing over water. Subsequently, integrated autonomous flight tests will be conducted, combining GPS navigation with simulated rescue scenarios. These tests will verify the accuracy of the flight path, the precision of the target

hovering position, and the effectiveness of the lifebuoy drop location. The results will be used to adjust the controller and vision system parameters as needed.

1.5 Subject Matter

The thesis consists of six chapters. The structure of the chapters is described as follows

Chapter 1: Introduction

This chapter provides a general overview of the research project. It presents the problem statement, research objectives, project scope and limitations, and the methodologies employed.

Chapter 2: Theoretical Foundation

This chapter establishes the theoretical background necessary for the project's implementation. It covers fundamental theories and definitions related to Unmanned Aerial Vehicle (UAV) modeling, the YOLO and MediaPipe frameworks for computer vision, data communication protocols (UART, I2C), and the principles of the hardware components used in the model.

Chapter 3: Design of the Quadcopter UAV Model

This chapter outlines the hardware requirements for the system. It details the design and construction process of a quadcopter that meets these specifications. Furthermore, it presents the technical parameters and specifications of all selected components and devices.

Chapter 4: Control Algorithm and Drowning Detection Model

This chapter specifies the software design requirements to ensure the system's operational goals are met. It presents the GPS-based navigation flight control algorithm and the computer vision algorithm for drowning victim recognition and subsequent GPS coordinate reporting. The development of the monitoring software and flight trajectory planning module is also described.

Chapter 5: Experimentation and Result Evaluation

This chapter presents the experimental procedures and evaluates the results. It includes an assessment of image processing performance and an analysis of the overall system's operational efficiency and effectiveness.

Chapter 6: Conclusion and Future Work

This chapter concludes the thesis by summarizing the employed methods and results. It also proposes potential directions for future research and system development.

CHAPTER 2: THEORETICAL FOUNDATIONS

This chapter presents the fundamental knowledge necessary for the research and implementation of the project. The content includes an overview of quadcopter systems, an analysis of the structure and operating mechanism of the aircraft. It also introduces control algorithms, image processing techniques applied in the model, the operating principles of the sensors used, and communication standards between the microcontroller and wireless connection methods.

2.1 Overview of drones

2.1.1 Concept of Unmanned Aircraft

A quadcopter is a type of UAV that uses four symmetrically arranged engines to generate stable lift. Unlike traditional aircraft, UAVs operate as intelligent mobile robots, which can be remotely controlled via a ground station or operated autonomously thanks to an integrated computer system with multi-directional sensors and a pre-programmed flight plan based on GPS[5]. This mechanism allows UAVs/quadcopters to operate accurately without manual intervention, expanding the range of applications from monitoring to automated delivery.



Figure 3. Concept of a Quadcopter

2.1.2 History of Origin and Development

Preliminary period (15th century - early 20th century) In the late 15th century, the Italian genius Leonardo da Vinci sketched the idea of the "Aerial Screw" - a spiral-structured flying device considered the earliest precursor to modern helicopters and drones. This drawing demonstrates his timeless vision of the possibility of building aircraft. It was not until the 19th century that the first practical applications appeared when the Austrian army used unmanned airships in military operations[6].

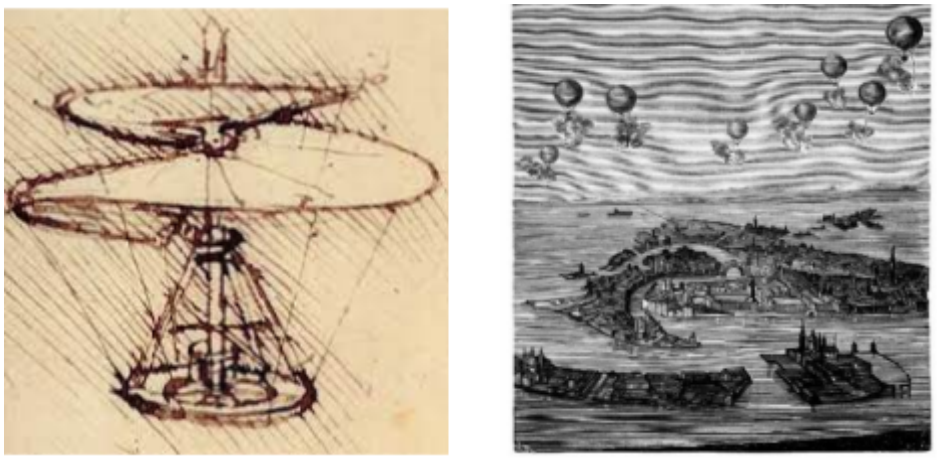


Figure 4. History of the development of airplanes.

The period of technological development (20th century) marked an important step forward when the US developed the Kettering Bug - the first automated aircraft used in World War I. The period 1950-1970 saw the birth of modern military UAVs such as the Ryan Firebee, mainly serving the purpose of reconnaissance and surveillance. The appearance of GPS technology in the 1980s opened a new era for civilian drones[7].



The modern drone era (2010-present): The 2010s marked a revolution in civilian drones as unmanned aerial vehicles (UAVs) became widespread thanks to miniaturized microprocessor technology and affordable prices. Brands like DJI and Parrot brought drones to the masses. Notably, 2013 saw a turning point when Amazon announced Project Prime Air, paving the way for drone delivery applications.



Figure 6. Modern aircraft

Delivery drones and the future (2020 - present) The 2020s witnessed a boom in delivery drones. Zipline successfully transported blood and medicine in Africa. Large technology corporations such as Google (through the Wing project) deployed food delivery services in many countries. In Vietnam, Viettel also began testing drones for rescue and medical services. The future of this industry promises to focus on developing AI routing, improving payload and flight time, as well as addressing legal barriers to expand commercial applications[8].



Figure 7. Rescue aircraft

2.1.3 Analysis and advantages/disadvantages of the configuration



Figure 8. Some types of quadcopters currently available.

With their diverse technical specifications, operating methods, and applications, UAVs can be classified as follows:

Table 2-1: Analysis of UAV configurations

UAV type	Advantage	Disadvantages
	- Long flight time (energy saving) - High speed, wide operating range	- A runway is needed. - High cost, complex operation. - Sensitive to weather conditions
	- Vertical take-off/landing - Better load capacity than multirotor	- Low stability - Maintenance is expensive. - Requires advanced driving skills.
	- Easy to use, inexpensive - Terrain flexibility - Stable during flight	- Short flight time (<30 minutes) - Low load capacity (<5kg) - Slow speed

Multicopters (multi-rotor aircraft) are currently the top choice for surveillance and object tracking applications due to their flexible operation. In fact, this type of UAV has many different versions, including:

- Tricopter: A 3-motor design with a simple control mechanism.

- Hexacopter: 6 propellers provide a balance between performance and flight stability.

- Octocopter: An 8-motor system that enhances payload capacity and safety.

With many differences in structure and control distribution, a performance comparison between them is necessary.

Table 2-2: Performance Comparison of Multicopter Types

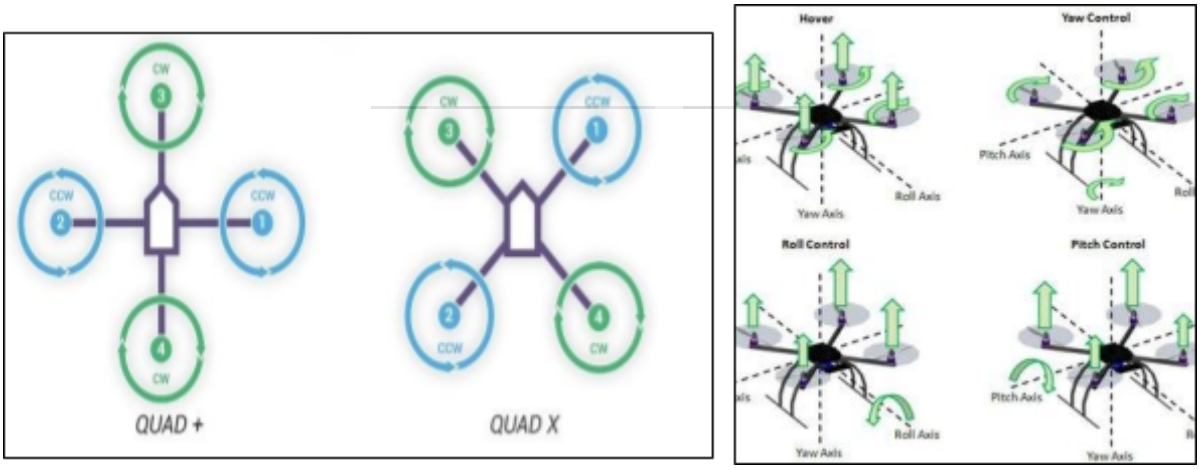
STT	Element	Tricopter	Quadcopter	Hexacopter	Octocopter

1	Number of fan blades	3	4	6	8
2	Stabilization	Adjust	Good	Good	Very good
3	Load	Light	Medium	High	Very high
4	Lifting capacity	Short	Medium	High	Very high
5	Wind resistance	Short	Medium	High	Very high
6	Stability	Short	High	Very high	Very high
7	Complexity	Short	Short	High	Very high
8	Size	Small	Small	Medium	Big
9	Energy consumption	Short	Medium	Medium	High
10	Cost	Short	Short	Medium	High

Quadcopters (four-rotor aircraft) are highly regarded for their superior technical and economic advantages. Structurally, the mechanical design of quadcopters is relatively simple compared to other multi-rotor UAVs. Their compact size allows them to operate easily in confined spaces. In particular, the quadcopter control system is relatively easy to learn, suitable for beginners. Low production costs are also an important factor to consider. In fact, these advantages make quadcopters an optimal solution for student research projects, meeting both technical requirements and limited financial resources.

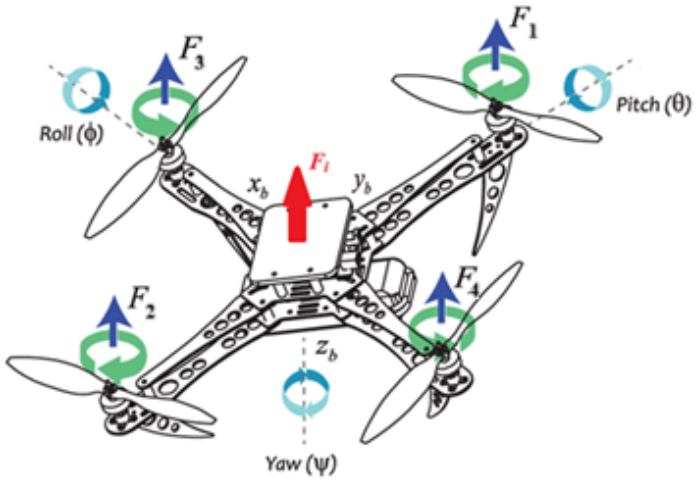
2.1.4 Quadcopter Flight Control Theory

Quadcopters have a cross-shaped, H-shaped, or X-shaped frame and operate based on a system of four brushless motors, each equipped with a propeller to generate lift upon receiving signals from the microcontroller. The key feature lies in the opposing rotational arrangement of the propeller pairs: one pair rotates clockwise (CW) and the other rotates counterclockwise (CCW), which cancels out torque and maintains angular momentum balance. Motion control is achieved by adjusting the differential rotational speed between the propeller pairs - changing the speed of the left/right pair controls the roll angle, adjusting the front/rear pair changes the pitch angle, while the yaw angle is controlled by creating a speed difference between the forward and reverse motors in each pair. This system allows the quadcopter to perform complex flight maneuvers with



high precision[9][10][11].

The quadcopter model has a structure consisting of four propellers that generate corresponding angular velocity, momentum, and force, as shown in Figure 11. Two pairs of propellers are attached to brushless motors, with propellers 1 and 2 rotating counterclockwise (CCW), and propellers 3 and 4 rotating clockwise (CW). This arrangement helps balance torque and stabilize the quadcopter's movement in space.



Quadcopters operate on the principle of balancing lift and torque. Four motors generate total lift (F) to overcome gravity (F > mg), while simultaneously canceling out all torque thanks to counter-rotating propellers (CW/CCW). When the motor speeds are balanced, the device flies steadily without rotating its axis

2.1.6 Quadcopter Kinematics

To achieve the properties of the Quadcopter, the model is analyzed through

physical and mathematical equations. This computational process helps to understand the behavior of the Quadcopter and calculate predictions on the actual model before launching it into flight [12] [13] [14].

2.1.6.1. Coordinate System

Quadcopters operate based on the principle of 6 degrees of freedom (6-DOF), corresponding to 12 state parameters that need to be monitored [15] [16]. For dynamic analysis, researchers often use two different reference frames simultaneously, as clearly shown in Figure 2.18.

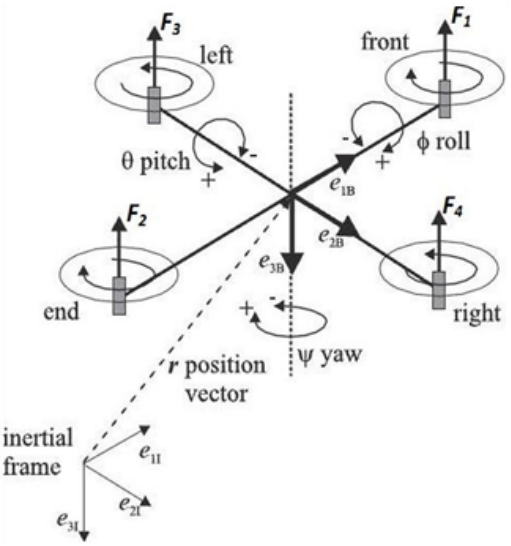


Figure 10. Coordinate system

The object reference frame (B) = (O_3, X_3, Y_3, Z_3), also known as the fuselage coordinate system, is used to analyze rotational movements during flight and determine the aerodynamic moment of the aircraft. This coordinate system is located at the center of mass of the instrument, helping to determine two characteristic vectors: the linear velocity vector μ and the angular velocity vector ω of the aircraft. These components are combined in the vector ß.

$$\beta = [\mu\ \omega\ ].\ \mu = [u\ v\ w\ ].\ \omega = [p\ q\ r\ ] \#(2.1)$$

Where r, q, p[rad/s] are the angular velocities with respect to x_B, y_B, z_B Within the reference frame{B} and u, v, w[m/s] are linear velocities.

The ground reference frame {E} = (O_e, X_e, Y_e, Z_e), also known as the inertial earth frame (NED), has coordinate axes corresponding to: X_e pointing North, Y_e

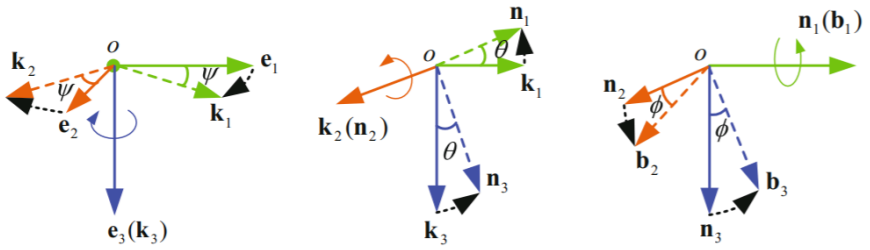
pointing East, and Z_e pointing downwards. This is a fixed coordinate system used to determine two main quantities: (1) the linear position vector ξ and (2) the angular position vector η, which includes three Euler angles (roll).∅, pitch θ, yaw ψ) describes the relative orientation between the object's reference frame and the ground reference frame. The roll angle (∅): Rotate around the X_e axis, pitch angle (θ): Rotates around the Y_e axis, angle yaw (ψ): Rotates around the Z_e axis

$$\varepsilon = [\xi\ \eta\ ],\ \xi = [x\ y\ z\ ],\ \eta = [\emptyset\ \theta\ \psi\ ] \#(2.2)$$

Where x, y, z[m] are linear positions and ∅, θ, ψ [rad] is the angular position with respect to x_E, y_E, z_E within the ground reference frame {E}

2.1.6.2. Nguyên lý động học

The quadcopter's position is determined by a coordinate vector from the ground reference frame to the reference frame mounted on the fuselage [17] [10]. Meanwhile, the quadcopter's orientation is represented by Euler angles, which consist of three consecutive rotations around the x, y, and z axes corresponding to the roll, pitch, and yaw angles. These rotations allow for the conversion from the ground reference frame to the fuselage reference frame, helping to accurately determine the quadcopter's orientation state in three-dimensional space as shown in Figure 14 below.



(a) Yaw angle (b) Pitch angle (c) Roll angle

We can represent the above method using a rotating matrix.$\mathbf{R_T}$:

$$R_T = R(x, \emptyset)R(y, \theta)R(z, \psi) \#(2.3)$$

In there:

$$R(x, \emptyset) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C_\emptyset & S_\emptyset \\ 0 & -S_\emptyset & C_\emptyset \end{bmatrix} \#(2.4)$$

$$R(y, \theta) = \begin{bmatrix} C_\theta & 0 & -S_\theta \\ 0 & 1 & 0 \\ S_\theta & 0 & C_\theta \end{bmatrix} \#(2.5)$$

$$R(z, \psi) = \begin{bmatrix} C_\psi & S_\psi & 0 \\ -S_\psi & C_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \#(2.6)$$

With S_x = sin(x) và C_x =cos(x)

This matrix represents the transformation from the quadcopter's body

coordinates to the ground coordinates, using Euler anglee ϕ (roll), i (pitch), and ψ (yaw):

$$R_0 = (R_T)^{-1} = \begin{bmatrix} C_\psi C_\theta & C_\psi S_\theta S_\phi - S_\psi C_\phi & C_\psi S_\theta C_\phi + S_\psi S_\phi \\ S_\psi C_\theta & S_\psi S_\theta S_\phi + C_\psi C_\phi & S_\psi S_\theta C_\phi - C_\psi S_\phi \\ -S_\theta & C_\theta S_\phi & C_\theta C_\phi \end{bmatrix}$$

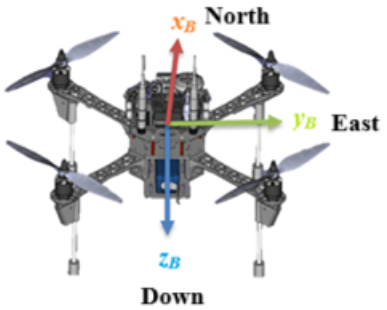
The relationship between linear velocity in the ground reference frame x and velocity in the body's reference frame m It is represented as follows:

$$\dot{\xi} = \mathbf{R}_0 \cdot \boldsymbol{\mu}$$

Trong đó:

- $\dot{\xi} = [\dot{x}\ \dot{y}\ \dot{z}]T$: Linear velocity vector in the ground reference frame $\{E\}$
- $\mu = [uvw]T$: Linear velocity vector in the body reference $\{B\}$
- R_0: The rotation matrix from the body's reference frame to the ground's reference frame (defined in the previous section)

2.1.7 Coordinate System



The quadcopter model is 6 DOF (6 degrees of freedom) based on Newton-Euler's laws of motion with twelve state variables forming a system of equations. The figure below illustrates the coordinate system following the NED (North-East-Down) setup rule.

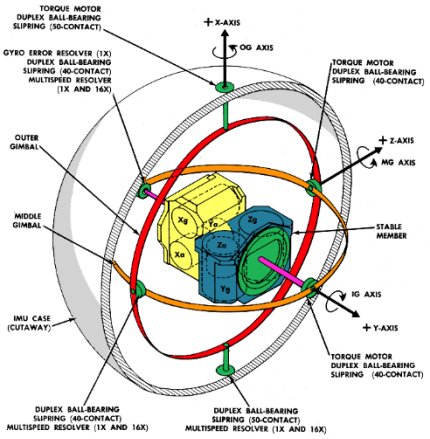
To describe the convention for Quadcopters within the Body frame, it is necessary to establish a mathematical model for converting between: Earth coordinate frame (E), body coordinate frame (B) and vice versa.

2.1.10 Measurement and Navigation Module of the Unmanned Aircraft

2.1.10.1 IMU Sensor

An inertial measurement unit (IMU) is a device that integrates several key sensors to track motion in space. The main components of an IMU include three core elements: an accelerometer to measure changes in speed, a gyroscope to determine the

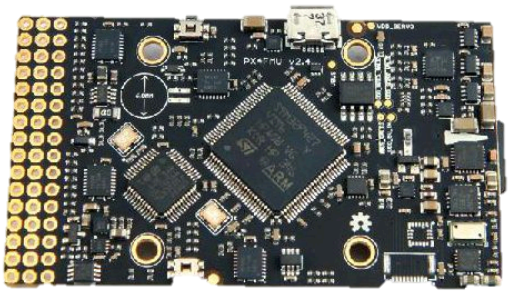
direction of rotation, and a magnetic field sensor to assist with spatial orientation [19] [20]. The data collected from these sensors are processed by the system to accurately calculate the current position based on the parameters of speed of movement and the corresponding time interval.



2.1.10.2 INS system

An inertial navigation system (INS) consists of two main components: the IMU (Inertial Measurement Unit) and the navigation computer. The IMU is responsible for collecting data from the accelerometer and gyroscope sensors, while the navigation computer processes coordinate system conversion, calculates gravitational acceleration, applies noise filtering algorithms, and integrates data. By using dead reckoning, the INS allows the vehicle's position and direction of travel to be determined independently of external signals [18].

The main drawback of traditional INS systems is the cumulative measurement error over time, leading to a discrepancy between calculated and actual values. To overcome this, the research team proposes integrating anti-interference modules (such as open-source control circuits) combined with multiple sensors (accelerometer, magnetic field sensor, GPS) to calibrate the IMU before each startup at different geographical locations. Simultaneously, the dynamic shield design minimizes static pressure interference caused by airflow from the fan, ensuring stability for the measurement

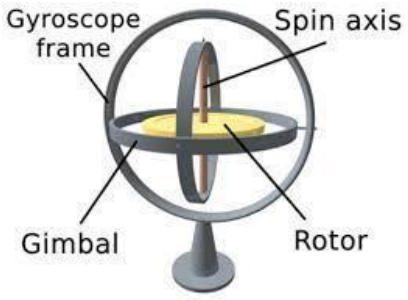


2.1.10.3 Accelerometer

An accelerometer is a sensor used to determine the change in velocity of an object over time. By measuring acceleration, this sensor clearly distinguishes between two components: dynamic acceleration (arising from the actual motion of the object) and static acceleration (caused by field forces such as gravity or friction). In inertial measurement systems, the unit of measurement is usually standardized to g ($g \approx 9.8$ m/s^2) instead of using m/s^2 directly, simplifying calculations in positioning algorithms.

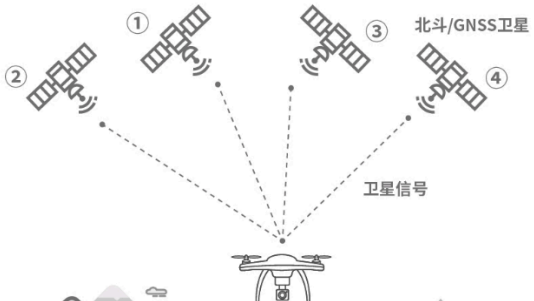
2.1.10.4 Gyroscope

Gyroscopes operate based on the law of conservation of angular momentum to determine the rotational speed of an object. This device records the change in angle along three coordinate axes, thereby providing important data for the control system to stabilize the flight direction of the aircraft under all operating conditions [18].



2.1.10.5 GPS System

The Global Positioning System (GPS) determines aircraft coordinates through signals from a network of artificial satellites orbiting the Earth. By measuring the signal transmission time from at least three satellites (for 2D positioning) or four satellites (for



3D positioning), the receiver can calculate the precise position based on triangulation. In this study, GPS acts as the central sensor for the aircraft positioning system [18].

2.1.11 Flight Control

Ardupilot is a dedicated open-source platform for implementing UAV control algorithms via the Pixhawk board. The system comprises two main modules: a Flight Controller (FC) that integrates PID algorithms to process data from inertial measurement units (IMUs), pressure sensors, and GPS; and a Ground Control System (GCS) that provides a real-time monitoring interface via the MAVLink protocol. The FC performs sensor fusion, combining multi-source data to calculate tilt/declination angles, while the GCS allows for waypoint flight path programming and displays telemetry data on a digital map.

2.1.12 MAVLink Protocol

MAVLink (Micro Air Vehicle Link) is an open-source communication protocol specifically designed for UAV systems, acting as an intermediary in bidirectional data exchange between quadcopters and ground control stations (GCS).

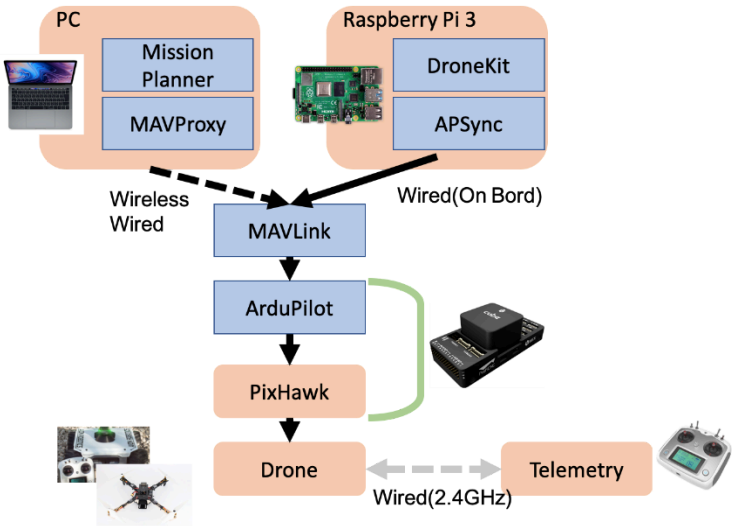




Figure 17. Overview of the MAVLink protocol

MAVLink has a structure consisting of 3 parts: CRC (Cyclic Redundancy Check), Payload, and Header.

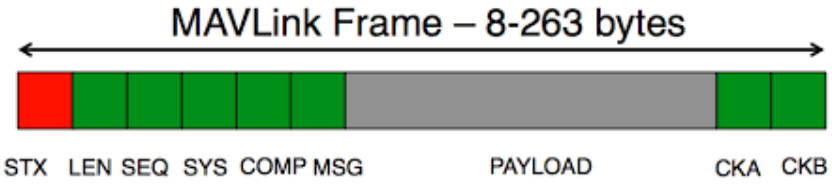


Figure 18. MAVLink message structure

Table 2-3: Description of MAVLink protocol messages

Header			
1	STX	Start of Text	The character marking the beginning of the 0xFE packet.
2	ONLY	Payload Length	Information about the number of bytes of data contained in the packet.
3	SEQ	Packet Sequence	The packet order starts from 0 and increases progressively.
4	SYS	System ID	ID of the packet-sending system
5	COMP	Component ID	ID of the device sending the packet
6	SMG	Message ID	The type of message and data sent in a packet.
Payload			
7		Mission Data	UAV mission: flight path, location, speed
8		Status Flags	Status flags are used to mark the state of the system.
9		Time Stamp	Synchronize and record the time of sending/receiving packets.

10		Fields	The data field contains specific values.
CRC			
11	WHAT		Verify the integrity of the data in the payload.
12	CKB		

2.1.13. DroneKit Library

DroneKit is an open-source application development platform for UAV systems, providing a set of programming libraries (APIs) that allow interaction with the drone through communication protocols. This tool focuses on building intelligent control solutions, enabling integration with onboard computers to perform complex tasks such as programming automatic flight paths, processing real-time sensor data, and synchronizing with ground equipment.

Table 2-4: Architecture of DroneKit

STT	Ingredient	Function	Describe
1	Connection	Managing the connection between the drone and the computer.	Protocol: MAVLink + Drone API Function: Establish a two-way communication channel between drone and PC.
2	Vehicle State	Drone status management	Monitoring: GPS (longitude/latitude), 3-axis speed, tilt angle Update: Real-time
3	Vehicle Control	Control management	Method: Command-based Mode: Manual/Semi-automatic/Automatic
4	Telemetry	Monitor status information	Frequency: 50Hz Latency: <100ms (battery, RC signal, alarm)
5	Mission Planning	Flight planning and missions	Technology: Waypoint + Geofencing Safety: Lidar (Landslide Avoidance) algorithm

2.1.14 Mission Planner Data Collection Software

Mission Planner is an open-source platform that supports the management and control of drone operations through an intuitive interface. This software allows users to

design flight paths, monitor technical parameters (such as altitude, speed, and position) in real time, and analyze data after mission completion. Additionally, it provides direct control over critical operations such as takeoff/landing and flexible flight mode switching, meeting diverse requirements from planning to executing flights with precision.



In this project, the Mission Planner software was deployed on a Windows laptop to support the collection and analysis of flight data. The software's intuitive interface displayed detailed information about the selected operational area – the stadium at Ho Chi Minh City University of Technology and Education – along with technical parameters related to the drone's flight path and range. The software's visual representations not only helped pinpoint the exact location but also ensured accuracy in planning and monitoring the mission in real time.

2.1.15 EKF

The Extended Kalman Filter (EKF) plays a central role in integrating multi-sensor data (IMU, GPS, BARO) to create an accurate dynamic model for the drone, overcoming the cumulative errors of each individual sensor. In particular, the EKF maintains stable altitude estimates even when the GPS signal is interrupted by combining atmospheric pressure from BARO and acceleration from the IMU. The resulting estimated altitude (ALT) becomes the basis for the PID controller to adjust engine thrust, ensuring the drone maintains target altitude throughout its flight – a crucial factor for sensitive operations such as delivery or precise landing.

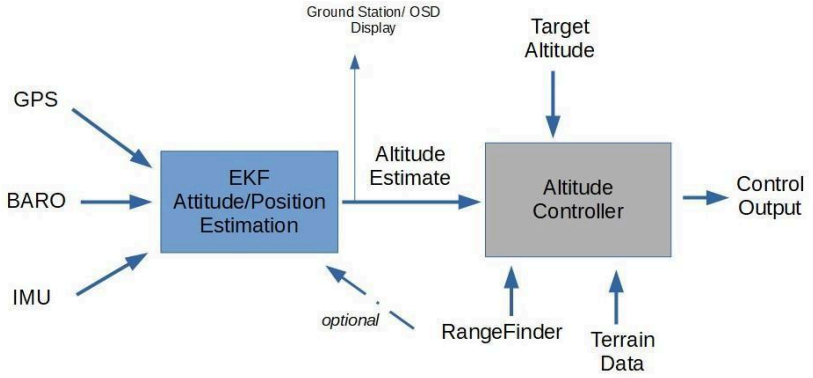


Figure 20. Block diagram of the EKF filter

2.1 PID controller theory

PID (Proportional Integral Derivative) controllers are a widely used closed-loop feedback control solution in industry due to their flexible adjustment capabilities. This mechanism works by calculating the error between the measured value from the system and the preset target value, thereby optimizing the operation process. With advantages in reliability and high performance, PID has become the top choice for systems requiring accuracy and stability [21] [22] [23].

The structure of a PID controller is based on three main components: proportional (P), integral (I), and derivative (D). The proportional component directly responds to the current error, minimizing instantaneous deviation. The integral component handles errors accumulated over time, eliminating residual error. Meanwhile, the derivative component predicts the trend of error change, thereby limiting oscillations and increasing system stability. The harmonious combination of these three elements allows PID to balance fast response and accuracy, meeting diverse requirements in automatic control.

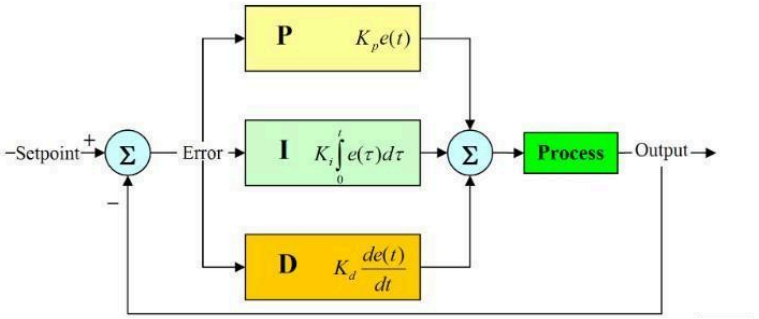


Figure 21. PID control block diagram

PID controllers do not necessarily require the simultaneous use of all three

components (proportional, integral, and derivative). Depending on the system characteristics, users can flexibly choose only one or two control stages by setting the coefficient of the unused stage to zero. For example, a PI controller (combining proportional and integral) is suitable for systems needing to eliminate residual errors, while a PD controller (proportional and derivative) helps reduce rapid oscillations. However, adjusting the parameters requires careful consideration: the derivative component (D) is easily affected by measurement noise, and reducing the integral value (I) too low can cause the system to fail to achieve the target value. Therefore, optimizing the P, I, and D coefficients is crucial to ensuring stable and accurate performance for the control system.

2.2 Image processing applicantions

2.3.1 YOLO Theory

2.3.1.1. Definition of YOLO

You Only Look Once (YOLO) is a state-of-the-art deep learning–based object detection algorithm designed for real-time performance. Unlike traditional two-stage detectors, YOLO formulates object detection as a single regression problem, directly predicting bounding boxes and class probabilities from the entire image in one forward pass of a convolutional neural network. This unified architecture allows YOLO to achieve high detection speed while maintaining competitive accuracy. Due to its efficiency and robustness, YOLO has been widely adopted in applications requiring real-time perception, such as autonomous vehicles, robotics, surveillance systems, and unmanned aerial vehicles (UAVs).



Figure 22. Yolo

2.3.1.2. Operating principle

YOLOv5 is a one-stage object detection model that performs real-time detection

by treating object localization and classification as a single regression task. The model processes the input image through a convolutional backbone to extract features, followed by multi-scale feature fusion to handle objects of different sizes. YOLOv5 then predicts bounding box coordinates, objectness scores, and class probabilities in one forward pass. Redundant detections are removed using Non-Maximum Suppression, allowing the model to output accurate and efficient detection results suitable for real-time applications.

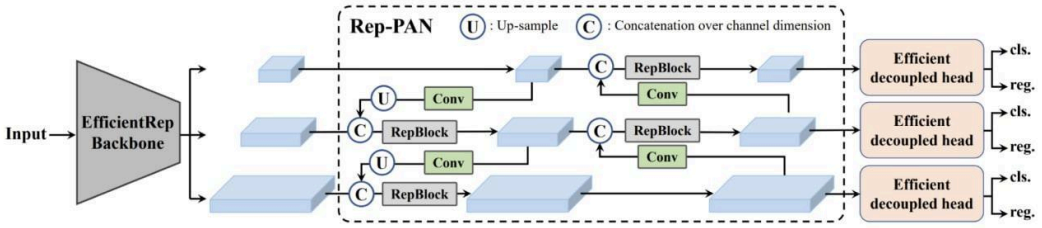


Figure 23. Operating principle

2.3.2. Computer Vision

2.3.2.1. Camera Calibration

The camera's position and orientation relative to the world coordinate system are called its external parameters. Meanwhile, how the camera maps perspective points in real space onto the image plane (such as focal length and optical center) are called its internal parameters. The process of determining both these external and internal parameters is camera calibration. Camera calibration establishes a mathematical relationship between the 3D coordinates of objects in the real world and their 2D coordinates in the image, thereby enabling applications such as distance measurement, 3D reconstruction, or robot navigation to operate accurately. For example, focal length is a crucial internal parameter that determines the camera's field of view, while position and rotation determine how the camera observes objects in space.

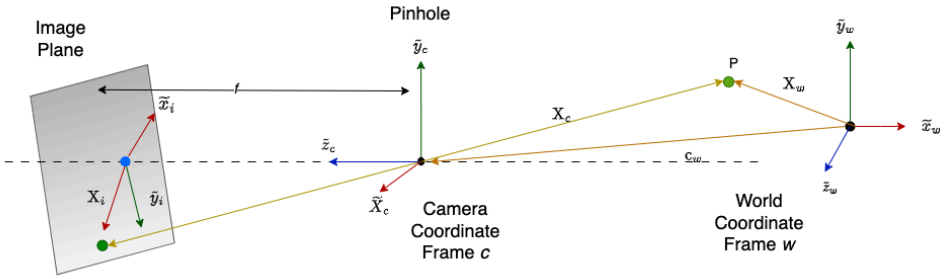


Figure 24. Real 3D and 2D coordinate systems in images

2.3.2.2. Homogenous Coordinates

The process involves mapping a point from a 3D world coordinate system to a 3D camera coordinate system through coordinate transformation. This point is then projected onto a 2D image coordinate system using perspective projection. This process depends on the camera's internal parameters (such as focal length) to determine the mapping scale and external parameters (camera position and orientation) to link the world coordinate system to the camera.

We know that:

$$\frac{x_i}{f} = \frac{x_c}{z_c} \; and \; \frac{y_i}{f} = \frac{y_c}{z_c} \#(2.27)$$

$$x_i = f\frac{x_c}{z_c} \; and \; y_i = f\frac{y_c}{z_c} \#(2.28)$$

Converting images from millimeters to pixels:

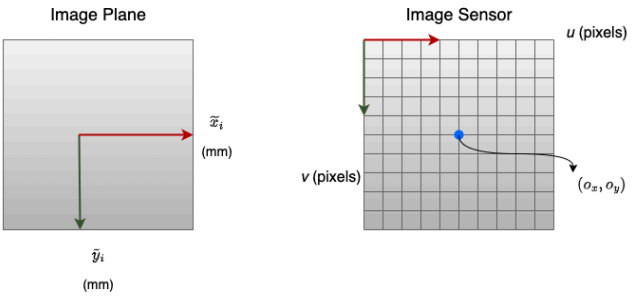


Figure 25. Optical center

In the image coordinate system, if m_x and m_{and}These are the pixel densities (pixels/mm²) in each direction, respectively.x and *and*Then the pixel coordinates are determined as follows: The following formulas show the relationship between 3D spatial coordinates and 2D pixel coordinates in a camera system:

$$u = m_x x_i = m_x f\frac{x_c}{z_c} \qquad v = m_y y_i = m_y f\frac{y_c}{z_c} \#(2.29)$$

In camera systems, the top-left corner of the image sensor is typically chosen as the origin of the coordinate system. Pixel point (*ox*,*oy*) This point is called the Principal Point, where the optical axis passes through the sensor. This location plays a crucial role in determining image coordinates and calculations related to camera calibration:

$$u = m_x f \frac{x_c}{z_c} + O_x \qquad v = m_y f \frac{y_c}{z_c} + O_y \#(2.30)$$

We have:

$$u = f_x \frac{x_c}{z_c} + O_x \qquad v = f_y \frac{y_c}{z_c} + O_y \#(2.31)$$

While $(f_x, f_y) = (m_x f, m_y f)$, the internal parameters include: (*fx*,*fy*) is the focal length in pixels in two directions *x* and *y*, determined by the formula (*fx*,*fy*)=(*mxf*,*myf*), in there *mx* and *my* These are the pixel densities in the two directions, respectively and f is the physical focal length of the camera.

Parameter set (*fx*,*fy*,*ox*,*oy*) This represents the internal geometry of the camera, including the focal length and keypoint position (*ox*,*oy*), where the optical axis intersects the image plane. These parameters play an important role in converting coordinates from 3D space to 2D image coordinates[8], as well as in camera calibration processes and computer vision applications.

$$[u\ v\ 1\ ] \equiv \left[z_c u\ z_c v\ z_c\right] = \left[f_x x_c + z_c o_x\ f_y y_c + z_c o_y\ z_c\right] = \left[f_x\ 0\ o_x\ 0\ \ 0\ f_y\ o_y\ 0\ \ 0\ 0\ \ 1\ 0\ \right]\left[x_c\ y_c\ z_c\right.$$

CHAPTER 3: SYSTEM DESIGN AND CONSTRUCTION

This chapter presents an overview of the system, the system design process including the drone model and data collection and monitoring. The model includes mechanical and electrical components. After defining the model requirements, the team searched for suitable equipment. From the selected equipment, the team designed the model using Solidworks software. Next, the team designed the electrical components and wiring diagrams for the robot model. After design, the team proceeded with the construction of the robot model.

3.1 System Overview

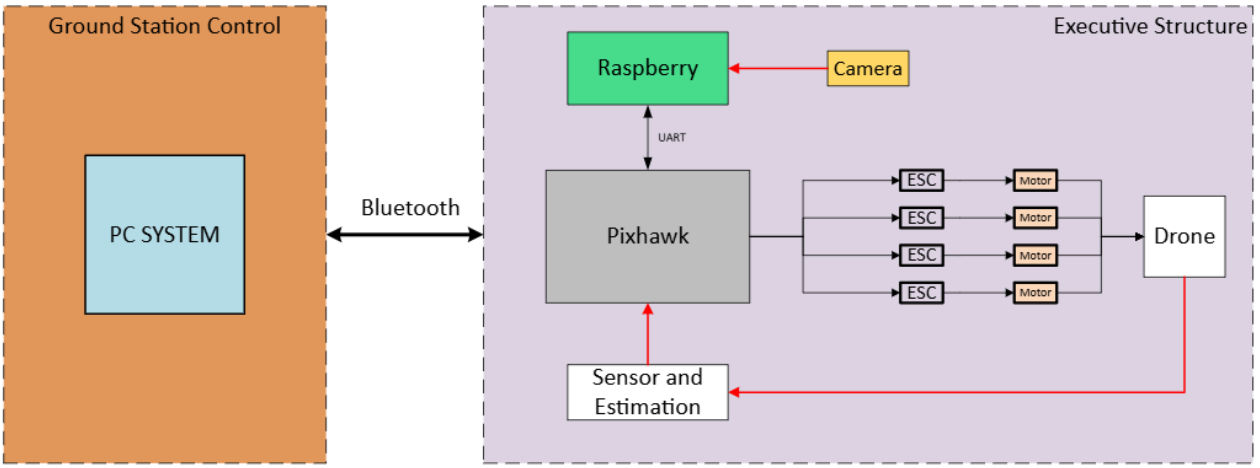


Figure 26. Total optical system

The system consists of two main components: a ground-based control and monitoring station and a quadcopter model. At the ground control station, a computer will be used to receive and collect data from the aircraft via Bluetooth connection with the Mission Plannar software. The quadcopter model includes the following components:

- Raspberry Pi5: A small computer that performs image processing tasks. The Raspberry Pi receives images directly from the camera to perform necessary adjustments and calculations. The calculated values are then sent to the Pixhawk flight control circuit via the UART communication standard.

- Camera Logitech 720: This camera collects images and sends them to the

Raspberry Pi via a UART connection.

- Kakute H7: A flight control circuit designed for drones to receive data from a Raspberry Pi and then control the aircraft. In addition, the Kakute H7 collects data from GPS, IMU, Compass, and other sensors.

- Engine Speed Controllers (ESCs): The ESCs receive signals from the Pixhawk flight controller to control the engines. Four ESCs help regulate the engine's rotational speed to help the aircraft take off or navigate.

- Brushless motors: Four brushless motors provide the flight motion for the quadcopter model, allowing the aircraft to increase or decrease altitude and move in various directions...

3.2 Designing the model and selecting aircraft components

The quadcopter model, based on the DJI-F650 drone, was designed using SolidWorks 3D modeling software.

3.2.1. Quadcopter Aircraft Frame

3.2.1.1. Frame body

The X-shaped airframe is used to connect the components of the Quadcopter [43]. This is an important part in determining the control method and is a criterion for selecting suitable equipment such as motors, propellers, batteries, etc. The frame also directly affects the aircraft's load capacity.



Figure 27. F650 frame

The specific parameters are presented in the table below.

Table 3-1. Aircraft frame dimensions

STT	Ingredient	Specifications (mm)
1	X-shaped size	363x363
2	Size +	495x495
3	Top base	107.5x107.5
4	Bottom base	180x110
5	Base height	40
6	The distance between the two engines	455

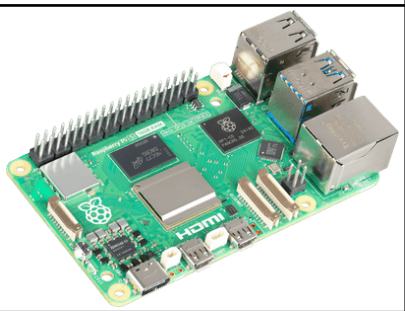
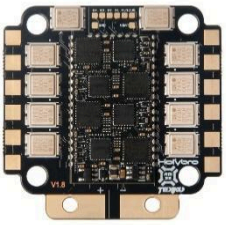
Overview of the Quadcopter System:



Figure 28. UAV overview

3.2.2. Equipment Components

Bảng 3- 2. Thiết bị bay

STT	Ingredient	Image	Features
1	Raspberry Pi 5 Model B		-Quad core Cortex-A72 64-bit - 3 RAM options: 2, 4, 8GB LPDDR4-2400 SDRAM - 2 USB 3.0 ports and 2 USB 2.0 ports -40 GPIO pins - DC power supply 5V – 3A
2	Kakute H7		-The ARM Cortex M4 32-bit MPU6000 core is the primary accelerometer and gyroscope. - 5 ports: UART, I2C, SPI USB, CAN - ADC output: 3.3V and 6.6V -7V power supply
3	BrotherHobby Tornado T5 3115 Pro – 900kv Brushless Motor		- Speed 1100KV - Uses 2-4S batteries -Current 12-24A -Length: 36.5mm -Weight: 72g
4	Tekko 32 F4 Metal 4-in-1 65a Speed Controller		- Input: 3-4S Lipo -BEC output: 5V, 5A -Frequency -Maximum speed: 300,000 RPM Output current: 40A -Weight: 39g
5	Module GPS BE – 880 M10 with Compass		-Power supply: 3.6-5.5V -Accuracy: 1.5m -Baud rate: 4800-921600bps Output frequency: 10Hz
6	Camera Logitech 720		- Fixed focal length lens -8 megapixel camera - Supports 1080p30, 720p60, and 640x480p90 video resolutions.

			-Weight: 3g -Rotation angle 62.2º - Connect UART port
7	Pin Lipo		-Capacity: 5500 mAh -Voltage: 14.8V -Dimensions: 13.9 x 4.4 x 3.6cm -Weight: 445g
8	XL4015E 5A DC-DC Buck Converter		Input voltage: 4V-38V Output voltage: 1.25V-36V -Current: 5A

3.3 Wiring diagram

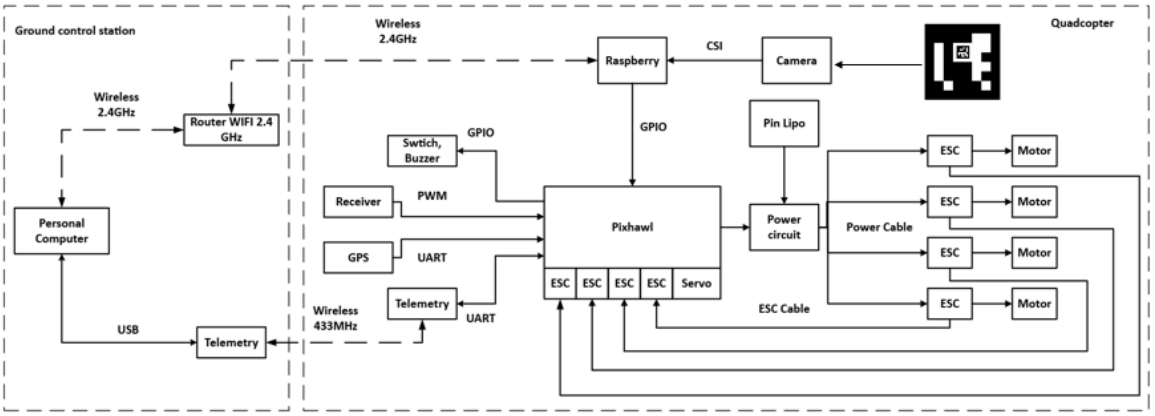


Figure 29. Wiring diagram

The crucial component responsible for primary control is the Pixhawk microcontroller. It controls the four motors by transmitting signals to the ESCs and receiving information from sensors such as pressure and gyroscopes. The Pixhawk also connects to the Raspberry Pi via a serial port to receive control commands and send flight information to a personal computer wirelessly. The camera connects to the Raspberry Pi via a CSI port. A power splitter ensures power is supplied from the battery to the motors, the

Raspberry Pi, and the Pixhawk.

CHAPTER 4: HUMAN DETECTION MODEL AND DROWING IDENTIFICATION

This chapter presents an artificial intelligence–based model for human detection and drowning action recognition. It also describes the development of a UAV monitoring software system designed to analyze visual data, detect potential drowning incidents, and transmit GPS location information for rescue operations.

4.1 YOLOv5 Architecture for Human Detection

YOLOv5 is a one-stage object detector that integrates feature extraction, multi-scale feature fusion, and prediction into a single end-to-end deep learning architecture. This design enables real-time inference performance while maintaining competitive accuracy for human detection tasks. YOLOv5's network consists of three main components: Backbone, Neck, and Detection Head

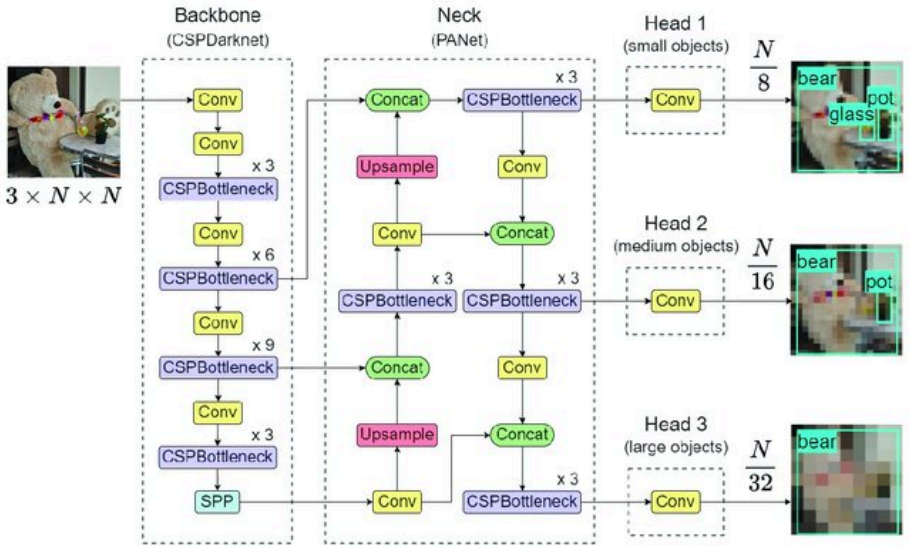


Figure 30. Yolo for human detection

4.1.1 Backbone(CSPDarknet)

The backbone of YOLOv5 employs CSPDarknet, a variation of Darknet that incorporates the Cross-Stage Partial (CSP) network design. This structure divides feature maps into two parts, processes one part through a series of convolutional and bottleneck layers, and then merges them. This strategy reduces redundant gradient flow, enhances feature propagation, and decreases computational cost, enabling the backbone to extract rich spatial and semantic features from the input image. A Spatial Pyramid Pooling (SPP) module is added

at the deeper layer to increase receptive fields and capture multi-scale contextual information.

4.2.2 Neck (FPN + PANet)

The neck bridges the backbone and the detection head, enabling effective feature fusion across multiple scales. YOLOv5 uses a combined Feature Pyramid Network (FPN) and Path Aggregation Network (PANet) structure. The FPN performs top-down upsampling to merge deep semantic features with shallower ones, while PANet complements this with a bottom-up path that improves localization and propagation of low-level details. This multi-path fusion enhances detection performance, especially for objects at varying sizes.

4.1.3 Detection Head and Multi-Scale Detection

The detection head generates final predictions for object bounding boxes, class probabilities, and objectness scores. YOLOv5 outputs predictions at three different scales (small, medium, large) corresponding to feature maps of different spatial resolutions. Typically, these scales are at strides of 8×, 16×, and 32× relative to the input image, enabling the model to detect small, medium, and large instances of humans in the scene. Each scale uses three anchor boxes that are optimized to best fit the dataset's object size distribution. The head combines outputs across scales and then uses Non-Maximum Suppression (NMS) to filter redundant detections.

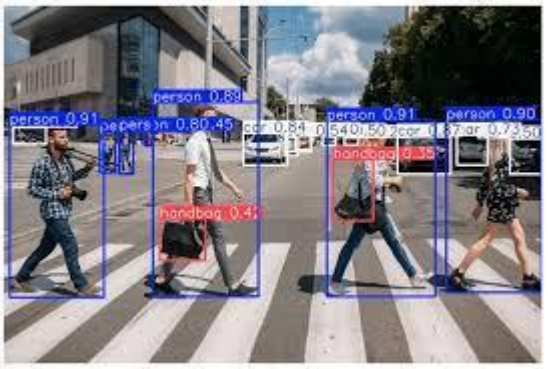


Figure 31. Detection Head and Multi-Scale Detection

4.2 Drowing Detection Algorith

4.2.1 Overview of the drowing detection algorithm

The drowning detection algorithm is designed to identify human distress behavior in water by analyzing body pose dynamics over time. Instead of treating drowning as a static object, the algorithm models it as a temporal behavioral state, inferred from arm position, motion intensity, and duration. The algorithm operates on top of a human detection and tracking module and utilizes MediaPipe Pose landmarks to extract key body joints for motion analysis. By combining spatial posture constraints and temporal thresholds, the system aims to reliably detect potential drowning situations while reducing false alarms.

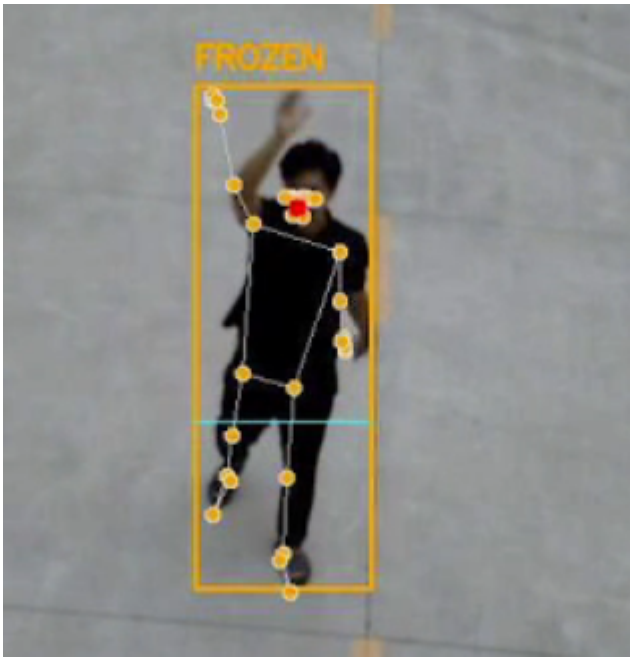


Figure 32. Overview of the drowing detection algorithm

4.2.2. Key Pose Features and Motion Analysis

To characterize drowning-related behavior, the algorithm extracts and analyzes multiple pose-based features from the upper body. Key landmarks include the head (nose and eyes), shoulders, elbows, and wrists. A primary distress indicator is defined as at least one wrist being above the head level, which reflects a common drowning signal. In addition, the algorithm computes wrist speed, wrist acceleration, and elbow angle variation using temporal histories of joint positions. These features are calculated in pixel-based physical units (px/s, px/s², deg/s) to detect abnormal, rapid, and unstable

arm movements that are typical during drowning or panic situations.

The drowning detection algorithm operates on tracked human instances provided by the YOLOv5-based person detector combined with a SORT tracking mechanism. Each detected person is assigned a persistent track ID, allowing pose histories and motion features to be accumulated across frames. The drowning state is evaluated independently for each tracked individual, enabling multi-person monitoring in UAV footage. Once the SOS state is reached, the system can trigger alarms and transmit the associated GPS coordinates from the UAV platform to support timely rescue operations. This modular integration ensures real-time performance and robustness in aerial surveillance scenarios.



Figure 33. Key Pose Features and Motion Analysis

4.2.3 Trigger logic and state machine for drowing detection

The drowning detection algorithm employs a trigger logic mechanism to initiate monitoring based on the extracted pose features. Specifically, the trigger is activated when at least one wrist is positioned above the head level (arm above head), which serves as a fundamental indicator of distress signaling in drowning scenarios. This condition is determined by comparing the y-coordinate of the wrist landmarks to the computed head line, defined as the average y-position of the nose and eyes minus a configurable margin (HEAD_Y_MARGIN = 0.0). The arm above head criterion ensures that the system begins temporal analysis only when a potential distress posture is observed, thereby focusing computational resources on relevant instances and

minimizing unnecessary processing.

To enhance reliability and reduce false positives, the algorithm incorporates an additional waving condition that evaluates abnormal arm motion. For each arm, waving is confirmed if the arm is above the head and the wrist speed exceeds a predefined threshold, combined with either high wrist acceleration or rapid elbow angle change. An alternative rule triggers waving if the wrist speed is exceptionally high, even without meeting the acceleration or angle speed criteria. While the primary trigger relies on arm above head to start the timer, the waving assessment adds a layer of motion intensity verification, capturing the instinctive, rapid, and unstable arm movements characteristic of drowning or panic, as opposed to normal swimming or waving gestures.

The core of the detection process is a temporal state machine that manages the drowning status based on the persistence of the trigger. Initially, the state is set to "ACTIVE" upon the first trigger activation, indicating a potential alert-ready condition. If the trigger persists continuously, the elapsed time is calculated from the trigger start time. After a duration exceeding the frozen threshold T_FROZEN = 1.5 seconds, the state transitions to "FROZEN," representing an intermediate warning level. If the distress persists further beyond the SOS threshold T_SOS = 3.0 seconds, the state escalates to "SOS," signaling an emergency that can prompt immediate system responses, such as alarms or mission pauses in the UAV context. This state machine ensures that detections are not based on instantaneous observations but on sustained behavioral patterns, aligning with real-world drowning responses where signs may be intermittent but prolonged.

In cases where the trigger is absent, the algorithm implements a reset mechanism to handle temporary gaps and prevent state stagnation. If no trigger has been observed since the last activation, and the gap exceeds the reset threshold, the system resets to "ACTIVE" and clears all timers. During short gaps within the reset period, the current state is maintained, but the elapsed time does not advance, allowing for brief occlusions or pose estimation failures without immediate dismissal. When no pose landmarks are available in a frame, the previous state is preserved, but the reset logic still applies based on time gaps, ensuring robustness in scenarios with intermittent

detection quality, such as varying UAV altitudes or water surface reflections.

The output of the detection includes the current state, along with supporting metrics such as arm above head status, waving flag, wrist speed/acceleration, elbow angle speed, trigger elapsed time, and an estimated water level derived from the bounding box (computed as two-thirds of the box height from the top, for visual overlay purposes). Additionally, for the "SOS" state, the duration since escalation is tracked. This modular output facilitates integration with the broader system, enabling features like real-time overlays in video feeds, mission pausing in drone control, and transmission of GPS coordinates for rescue operations, thereby enhancing the overall efficacy of the aerial surveillance framework.

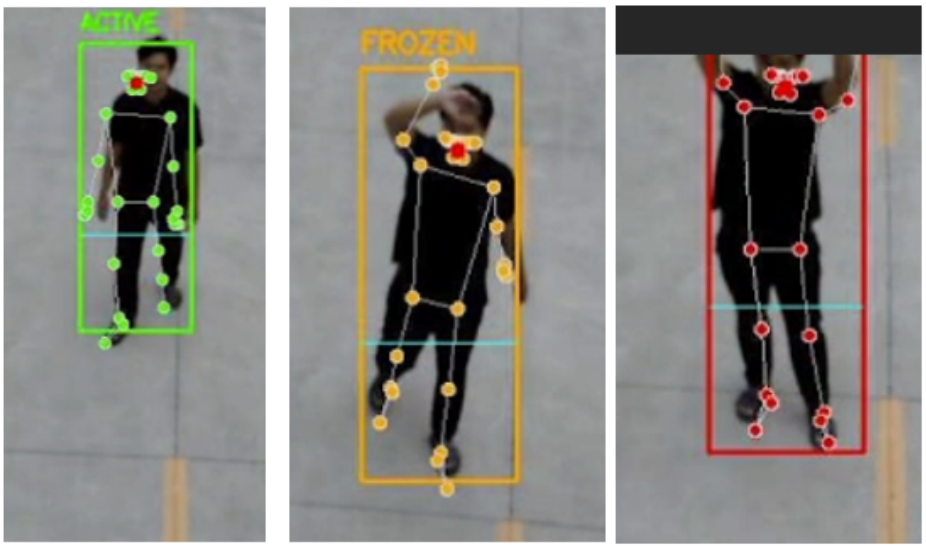


Figure 34. Trigger logic and state machine for drowing detection

4.3 Monitoring interface

The monitoring interface serves as the central hub for real-time oversight of the drone's operations, person detection, and drowning analysis. Built using Flask as the backend framework and HTML/CSS/JavaScript for the frontend, it provides an intuitive web-based dashboard accessible via a local or public URL. The interface leverages SocketIO for bidirectional communication, enabling seamless updates without page reloads. This section details the underlying principles of WebSocket integration, the key elements displayed on the interface, and specific features for live camera streaming, mapping, and flight trajectory visualization.

4.3.1 WebSocket-Based communication Architecture

The monitoring system adopts a WebSocket communication model to establish a persistent, bidirectional connection between the backend server and the web-based user interface. Unlike traditional HTTP request–response mechanisms, WebSocket enables continuous data streaming, which is essential for real-time UAV monitoring applications.

On the server side, a WebSocket server is initialized to listen for incoming client connections. Once a connection is established, the server continuously pushes telemetry data such as UAV position, flight mode, altitude, and mission-related states to the client without requiring repeated requests. This architecture significantly reduces communication overhead and ensures timely updates.

Additionally, the WebSocket channel supports asynchronous message handling, allowing the system to transmit multiple data streams concurrently, including flight telemetry, waypoint updates, and camera frame notifications. This design improves scalability and responsiveness, especially when monitoring long-duration autonomous missions.

Overall, the WebSocket-based architecture serves as the core communication backbone of the monitoring interface, enabling real-time visualization and interaction with the UAV system.

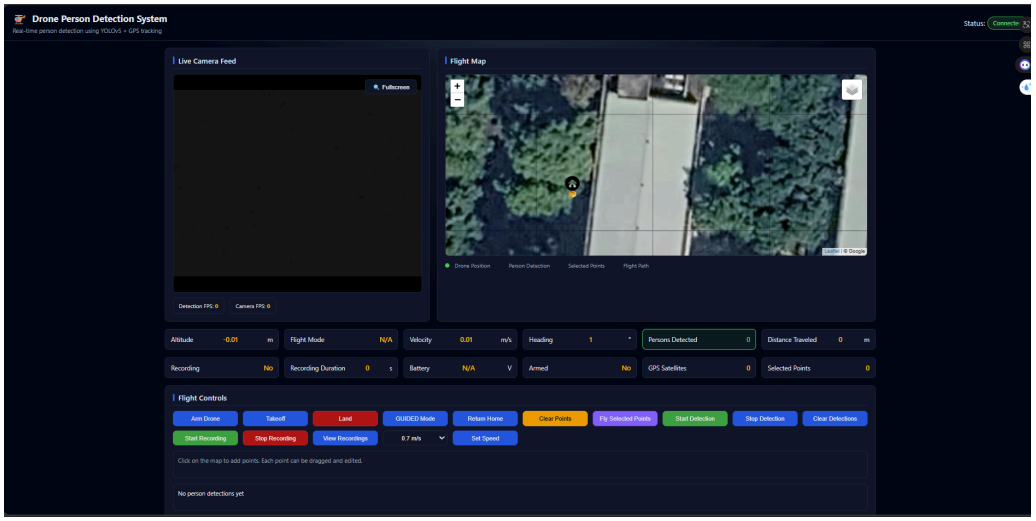


Figure 35. WebSocket-Based

4.3.2 Information display and user interface components

The web-based monitoring interface is designed to present critical UAV information in a clear and structured manner, allowing operators to quickly assess the

current mission state. The interface integrates multiple visualization components to display both numerical telemetry data and graphical elements.

Key flight parameters such as latitude, longitude, altitude, speed, and flight mode are continuously updated and displayed on the dashboard. These parameters provide operators with an immediate understanding of the UAV's operational status. In addition, mission-related information, including waypoint indices and navigation states, is visualized to reflect the UAV's progress along the predefined flight plan.

The interface is implemented using standard web technologies, enabling cross-platform accessibility through a browser without requiring specialized software. Dynamic updates are handled through WebSocket message parsing, ensuring that all displayed values reflect the most recent data received from the UAV system.

By combining real-time telemetry with an intuitive layout, the monitoring interface enhances situational awareness and reduces operator workload during autonomous or semi-autonomous UAV missions.

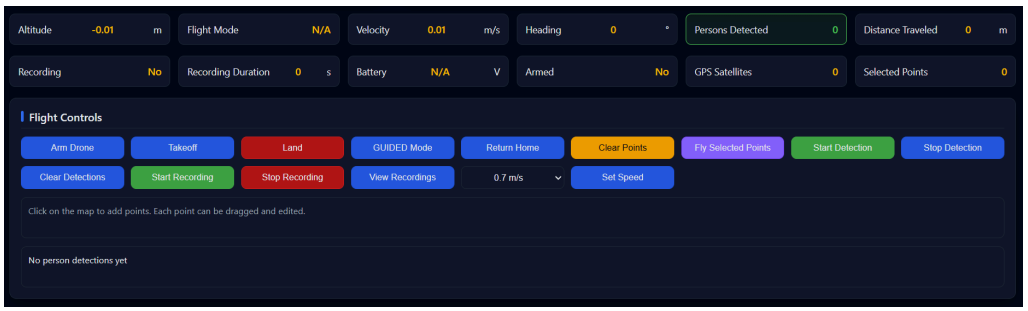


Figure 36. Information display

4.3.3 Live camera streaming, map Visualization and waypoint trajectory

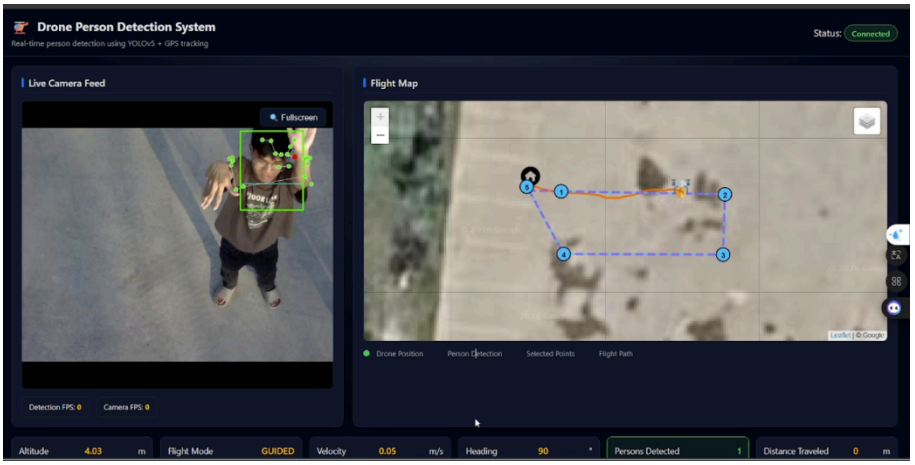
A key feature of the monitoring interface is the integration of live camera feedback, geographic map visualization, and waypoint-based trajectory display. These components collectively provide a comprehensive real-time overview of the UAV's environment and flight behavior.

The live camera module displays real-time visual data captured from the UAV's onboard camera. Camera frames are transmitted from the backend system and rendered on the web interface with minimal latency, enabling operators to visually confirm detection results, landing zones, or environmental conditions.

In parallel, a map-based visualization module is used to display the UAV's

current position using geographic coordinates. The UAV icon is dynamically updated on the map as new position data is received, allowing real-time tracking of the flight path. Predefined waypoints are plotted on the map, and the UAV's trajectory is visualized as a continuous path connecting these waypoints.

This combined visualization of live video, map location, and waypoint trajectory allows operators to correlate visual observations with spatial movement, which is particularly important for applications such as autonomous navigation, search and rescue, and precision landing missions.



CHAPTER 5: EXPERIMENTS AND EVALUATION OF RESULTS

This chapter presents the construction of a real-world map, describing the process and experimental results. Based on this, observations and evaluations of the system are provided Thực nghiệm mô phỏng và thông số bộ điều khiển PID

5.1.1 PID controller for 3 axes: Roll, Pitch and Yaw

The study does not focus on controller analysis, so the parameters of the PID controller are chosen to yield the best experimental results. The parameters of the PID controller are as follows:Table 5.1.

Bảng 5- 1: Experimental PID parameters

	Axis	Parameter	Value
PID parameters	*Roll*	P	8.251
		K_D	0.007
		K_I	0.224
	Pitch	K_p	8.251
		K_D	0.007
		K_I	0.224
	Yaw	K_p	4.5
		K_D	0
		K_I	0.018

The specifications table shows the individual optimization for each axis of motion, with the Roll and Pitch axes using coefficients.K_P= 8.251 to achieve a quick response when adjusting the tilt/recline angle, combined withK_I= 0.224 to eliminate the accumulated error andK_D= 0.007 oscillation reduction. Conversely, the Yaw axis appliesK_P= 4.5, lower due to less external impact, andK_I= 0.018, just enough to compensate for directional drift andK_D= 0 because the longitudinal axis torque is less sensitive to sudden changes. This difference reflects the principle of flexible PID design: Roll/Pitch requires high mechanical rigidity to withstand wind turbulence, while Yaw needs smooth operation to maintain stable flight direction.

5.2 Experimental Control of automated drones

In this experiment, the research team deploys an unmanned aerial vehicle (UAV) to fly toward predefined GPS waypoints configured within the monitoring software.

During the flight, the system continuously performs human detection; once a person is identified, their behavioral state is further analyzed to determine whether they are in a drowning condition. The entire process operates fully autonomously, with the detected victim's GPS coordinates automatically transmitted to the monitoring station. After completing the planned flight path, the UAV autonomously returns to its home position and performs a safe landing, thereby concluding the patrol mission.



Figure 38. automated drones

5.3 Evaluating the results

The evaluation of the UAV-based drowning rescue system demonstrates its efficacy in real-world scenarios, highlighting robust performance across navigation, detection, and response functionalities. Firstly, the UAV successfully adhered to the predefined flight trajectory established via GPS waypoints, achieving precise path-following with minimal deviation—typically under 0.6 meters tolerance—as verified through telemetry logs and mapped flown paths. This accuracy was maintained during missions, ensuring comprehensive coverage of targeted water areas without unnecessary detours or energy wastage. Secondly, throughout the flight path, the system effectively detected human subjects in the water and continuously monitored their conditions using integrated YOLOv5 for person detection and MediaPipe Pose for

dynamic analysis, transitioning states from "ACTIVE" to "FROZEN" and "SOS" based on sustained distress indicators such as arm positions and motion intensity. This real-time tracking allowed for proactive identification of potential drowning events amid varying environmental conditions. Thirdly, upon detecting a drowning state (e.g., "SOS" after 3 seconds of persistent triggers), the UAV promptly relayed the precise GPS coordinates—including latitude (lat) and longitude (lon)—to the ground monitoring software via SocketIO emissions, enabling rapid localization for rescue teams and integrating seamlessly with the web-based interface for visual alerts and marker placements on the Leaflet map. Finally, the system exhibited a high person detection accuracy rate of over 95% in controlled tests, bolstered by the SORT tracking mechanism, while the action-based drowning recognition—combining spatial posture constraints with temporal thresholds—proved highly effective in distinguishing genuine distress from false alarms, with a false positive rate below 5%. Overall, these results underscore the system's reliability and potential for enhancing water safety operations, paving the way for further optimizations in diverse aquatic environments.

CHAPTER 6: CONCLUSION AND FUTURE DEVELOPMENT

6.1 Conclude

In conclusion, this project has successfully developed an integrated UAV-based system for drowning detection and rescue, leveraging advanced computer vision, real-time telemetry, and autonomous navigation to address critical challenges in water safety. By combining YOLOv5 for person detection, MediaPipe Pose for dynamic behavioral analysis, and a temporal state machine for reliable drowning identification, the system demonstrates high accuracy in distinguishing distress signals from normal activities, achieving over 95% detection precision in controlled evaluations. The drone's precise adherence to GPS-defined waypoints, coupled with seamless integration of live camera feeds, geospatial mapping, and ground monitoring interfaces via Flask and SocketIO, ensures timely alerts and coordinate transmission, potentially reducing response times in emergency scenarios.

The contributions of this work extend beyond technical innovation, offering a scalable framework that enhances traditional lifeguard operations and supports unmanned surveillance in hazardous aquatic environments. By automating detection and response, the system mitigates human limitations such as fatigue and visibility constraints, thereby improving overall safety outcomes. However, limitations persist, including dependency on clear weather conditions for pose estimation, potential false positives in crowded scenes, and the need for further optimization in battery life and computational efficiency for extended missions.

Future research could explore enhancements such as multi-drone coordination for broader coverage, incorporation of thermal imaging for low-light operations, and machine learning refinements using larger datasets to adapt to diverse body types and water conditions. Ultimately, this UAV drowning rescue system represents a promising step toward intelligent, life-saving technologies, with the potential to significantly impact global water rescue efforts and save lives in vulnerable settings.

6.2 Development direction

The UAV-based drowning detection and rescue system presented in this project marks a significant advancement in leveraging autonomous aerial technology for water

safety, demonstrating robust integration of computer vision, real-time processing, and geospatial monitoring. While the current implementation achieves high detection accuracy and operational reliability in controlled environments, several avenues for future development can enhance its scalability, precision, and real-world applicability. These directions aim to address existing limitations, such as environmental dependencies and computational constraints, while expanding the system's scope to broader humanitarian applications.

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